

2023-Ph-H1-Q11

Section: Electricity

Topic: Capacitors

A $200\text{ }\mu\text{F}$ capacitor is charged to 12 V . What is the energy stored?

Answer: C

(0.014 J)

Guidance for Students:

Use energy formula:

$$E = \frac{1}{2}CV^2$$

Convert capacitance:

$$C = 200 \times 10^{-6} = 0.0002\text{ F}$$

So:

$$E = \frac{1}{2} \times 0.0002 \times 12^2 = 0.0144\text{ J} \approx 0.014\text{ J}$$

Revision Tips:

- Always convert μF to F .
- Energy stored in a capacitor:

$$E = \frac{1}{2}CV^2$$

- Check units: volts, farads, joules.